Training and Testing Sets Documentation

Project: Customer Churn Analysis for Telecommunications Company
Team: Team 3
Stage: Stage 2 – Data Preparation and Clustering Analysis
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1. Purpose

As part of the Stage 2 data preparation process, the preprocessed customer dataset was divided into separate training and testing sets.

The purpose of creating separate datasets is to support model development and validation. The training dataset is used to train the predictive model, while the testing dataset provides separate customer records that can be used to evaluate how well the trained model performs on unseen data.

2. Prepared Dataset

After the data cleaning and preprocessing process, the dataset contained a total of **6,741 customer records**.

During preprocessing:

- Missing values were checked.
- Duplicate records were removed.
- Categorical variables were converted into numerical representations.
- The target variable Churn was encoded into numerical values.
- The dataset was prepared for subsequent predictive modelling.

The predictive dataset contained **10 input features**, with Churn used as the target variable.

3. Training and Testing Split

The preprocessed dataset was divided using an **80/20 train-test split**.

The resulting datasets contained:

- **Training set:** 5,392 customer records
- **Testing set:** 1,349 customer records
- **Total records:** 6,741
- **Number of predictive features:** 10
- **Target variable:** `Churn`

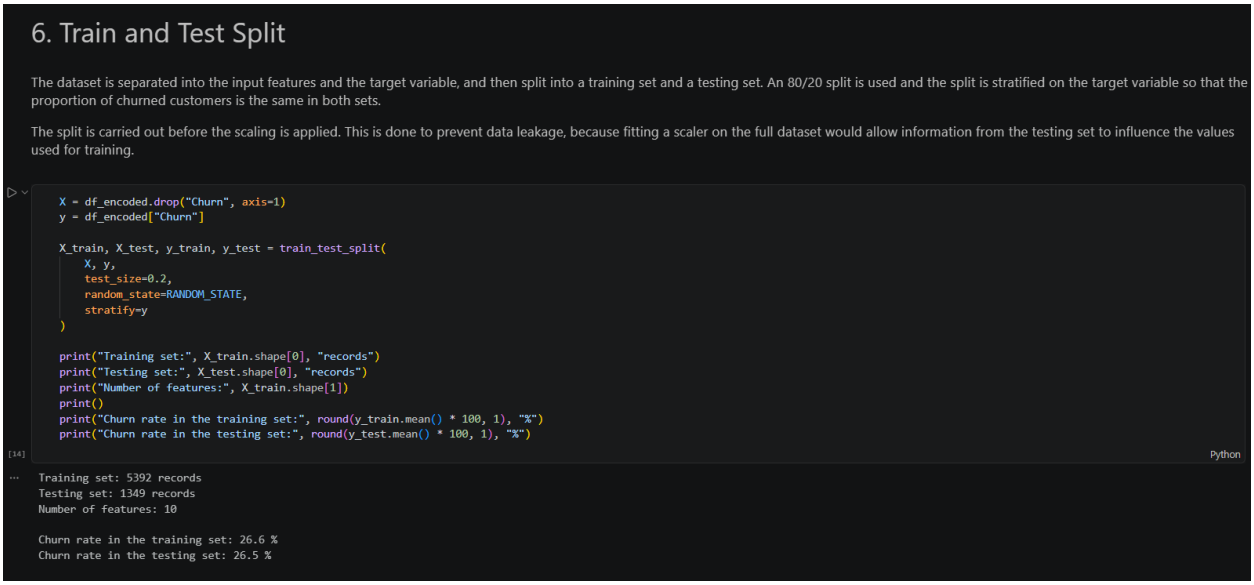
A `random_state` value of **42** was used so that the same training and testing split can be reproduced when the code is run again.

The `stratify=y` parameter was also used during the split. This helps maintain a similar proportion of churn and non-churn customers in both the training and testing datasets.

The churn rate was approximately:

- **Training dataset:** 26.6%
- **Testing dataset:** 26.5%

Evidence from Data Preparation Notebook



4. Composition of the Training and Testing Sets

The input features (X) were separated from the target variable (y) before the train-test split.

The target variable was:

`Churn`

The following **10 predictive features** were included:

1. `gender`
2. `SeniorCitizen`
3. `Dependents`
4. `tenure`
5. `PhoneService`
6. `MultipleLines`
7. `MonthlyCharges`
8. `InternetService_Fiber optic`
9. `Contract_One year`
10. `Contract_Two year`

The target variable Churn was stored separately from these predictive features.

This structure allows the predictive model to learn relationships between customer characteristics and the customer's churn outcome.

5. Training Set

The training dataset contains **5,392 customer records** and is used for model training.

The training data was exported into two separate files:

`ATS_train_features_v1.csv`

This file contains the **10 predictive input features** for the training customers.

`ATS_train_target_v1.csv`

This file contains the corresponding **Churn target values** for the training customers.

Separating the features and target variable provides a clear structure for subsequent predictive modelling.

6. Testing Set

The testing dataset contains **1,349 customer records**.

The testing data was exported into two separate files:

`ATS_test_features_v1.csv`

This file contains the **10 predictive input features** for the testing customers.

`ATS_test_target_v1.csv`

This file contains the corresponding **Churn target values** for the testing customers.

The testing set is kept separate from the training set so that it can later be used to evaluate the predictive model on customer records that were not used for model training.

7. Data Scaling

The numerical features `tenure` and `MonthlyCharges` were standardised using **StandardScaler**.

For predictive modelling, the scaler was fitted only on the training dataset and then applied to the testing dataset. This approach helps prevent information from the testing data from influencing the model-training process.

Detailed information about the scaling process, including the relevant code and evidence, is provided separately in:

`Team 3 -Scaling Techniques Documentation.pdf`

8. Output Files

The following training and testing files were generated during the data preparation process:

Training files

- `ATS_train_features_v1.csv`
- `ATS_train_target_v1.csv`

Testing files

- `ATS_test_features_v1.csv`
- `ATS_test_target_v1.csv`

These files are stored within the `Data_Preparation` folder as part of the Stage 2 submission.

9. Summary

The preprocessed customer dataset containing **6,741 records** was successfully divided into separate training and testing datasets using an **80/20 stratified split**.

The training dataset contains **5,392 records**, while the testing dataset contains **1,349 records**. Both datasets contain **10 predictive features**, with Churn maintained separately as the target variable.

The use of separate training and testing sets provides an appropriate structure for predictive model development and subsequent validation while maintaining a similar churn distribution across both datasets.